School_____ class: _____

NORTHERN EDUCATION DIVISION

2020 MALAWI SCHOOL CERTIFICATE OF EDUCTION MOCK EXAMINATIONS

AGRICULTURE

(100 MARKS)

Subject Number: MO12/I

Date: **Monday, 23rd March**

Time allowed: **2 hours**

(8:00 – 10:00 am)

PAPER I

Instructions

1. This page contains **11** printed pages. **Please check.**
2. This paper has two sections, A and B. Section A has **ten questions** and section B, **three** questions.
3. Answer **all** questions in the spaces provided.
4. Write your **name, form** and **school** on top of all pages.
5. In the table provided on this page, tick against the question number you have answered.
6. At the end of the examination, hand in your paper to the invigilator

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
10			
11			
12			
13			

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Section A (70 marks)

Answer all questions in this section

1. a. List any **two** exotic breeds of goats.

_____

_____

(2 marks)

b. Describe **one** way in which the following characteristics of a pig house increases production.

i. a warm house in winter

_____

_____

(2 marks)

ii. wallowing area.

_____

_____

(2 marks)

c. Explain any **two** ways in which soil temperature influences crop production

_____

_____

_____

_____

(4 marks)

d. List any **two** cultivated species of mushroom.

_____

_____

(2 marks)

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Figure 1. is a diagram showing irrigation systems. Use it to answer the questions that follow.

A B

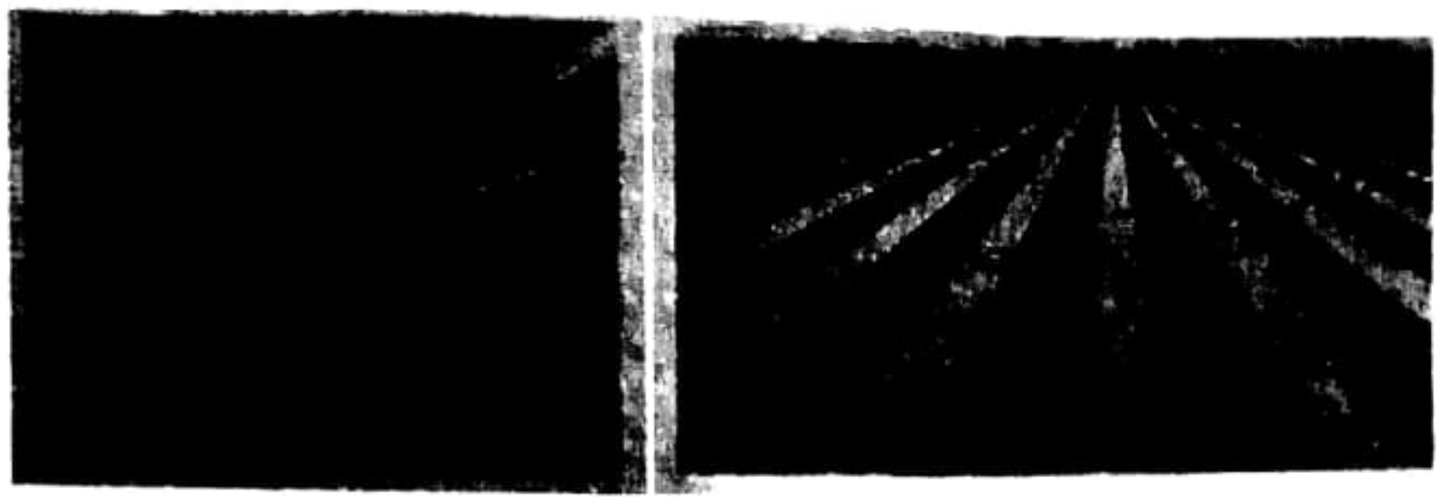

Figure 1

a. Identify the types of irrigation systems labelled A and B.

A _____ (1 mark)

B _____ (1 mark)

b. Explain any **two** problems associated with irrigation system B.

_____

_____

_____

_____

(4 marks)

c. Explain any **one** way in which the irrigation systems can ensure food security in Malawi.

_____

_____

(2 marks)

3. a. Explain **one** way in which staking can ensure good growth of tomatoes.

_____

_____

(2 marks)

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b. State any **two** roles performed by the itinerant trader in a marketing channel

_____

_____

(2 marks)

c. A farmer borrowed K200,000 from a bank at an interest rate of 10% per annum. He settled the loan in a year. Calculate the total amount of loan he paid.

(4 marks)

4. a. Explain any **one** way how each of the following mitigation measures deal with climate change in agriculture.

i. Afforestation

_____

_____

(2 marks)

ii. Rain water harvesting

_____

_____

(2 marks)

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b. State **two** ways in which top soil is important.

_____

_____

_____

(2 marks)

5. **Table 1** is a diagram showing an experimental design on fertilizer rates. Use it to answer the questions that follows

Plot	**Block I**	**Block II**	**Block III**
1	0kg/ha	50kg/ha	150kg/ha
2	50kg/ha	150kg/ha	100kg/ha
3	150kg/h	100kg/ha	0kg/ha
4	100kg/ha	0kg/ha	50kg/ha

Table 1

a. Name the design of the experiment

_____

(1 mark)

b. State the following:

i. control treatment

_____

(1 mark)

ii. Title of the experiment

_____

(1 mark)

c. Give any **two** factors to consider when designing the experiment.

_____

_____

(2 marks)

d. Explain **one** way how the experiment can ensure food security to the growing Malawian population.

_____

_____

(2 marks)

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6. **Figure** 2 below is a diagram showing a cattle management practice carried out in a crush. Use it to answer the questions that follow.

Figure 2

a, Identify the livestock management practice shown in **figure 2**.

_____ (1 mark)

b. Explain any **one** way in which the management practice in **figure 2** can reduce cost of production.

_____

_____

_____

(2 marks)

c. Explain any **one** way how tape worms can reduce beef production.

_____

_____

_____

(2 marks)

7. a. State any **two** ways of improving trading of agricultural commodities at community level.

_____

_____

_____

(2 marks)

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b. Describe any **two** principles followed when forming of an agricultural cooperative

_____

_____

_____

_____

_____

(4 marks)

c. A farmer intends to sow *Rhodes grass* on his pasture land. He had the following information.

- Seed size = 400,000 seeds/kg
- Expected plant population =1200,000 plant or seed
- Purity % = 80%
- Germination % = 30%

Calculate seed rate

(3 marks)

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8. a. Explain any **two** ways in which processing of crops is important.

_____

_____

_____

_____

(4 marks)

b. Most women in Malawi use simple farm tools. Explain **one** way how this gender bias leads to low agricultural production.

_____

_____

(2marks)

9. a. A farmer intends to grow **pasture** and rear **beef cattle** on a piece of land. Use the information to answer questions that follow.

i. Name the type of enterprise combination to be followed by the farmer.

_____ (1 mark)

ii. Explain **one** way in which the type of enterprise mentioned in **9.a.i.** can increase pasture production.

_____

_____

_____

(2 marks)

b. State any **two** ways in which farmers can reduce the effects of risks and uncertainties on a farm.

_____

_____

(2 marks)

10. a. Differentiate break-even budget and partial budget

_____

_____

(2 marks)

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b. Give any **two** uses of partial budget

(2 marks)

Section B (30 marks)

Answer **all** the **three** questions this section.

11. Describe any **five** factors that affect seed rate of pasture.

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12. Describe any **five** ways of dealing with gender biases in agricultural technology. (10 marks)

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